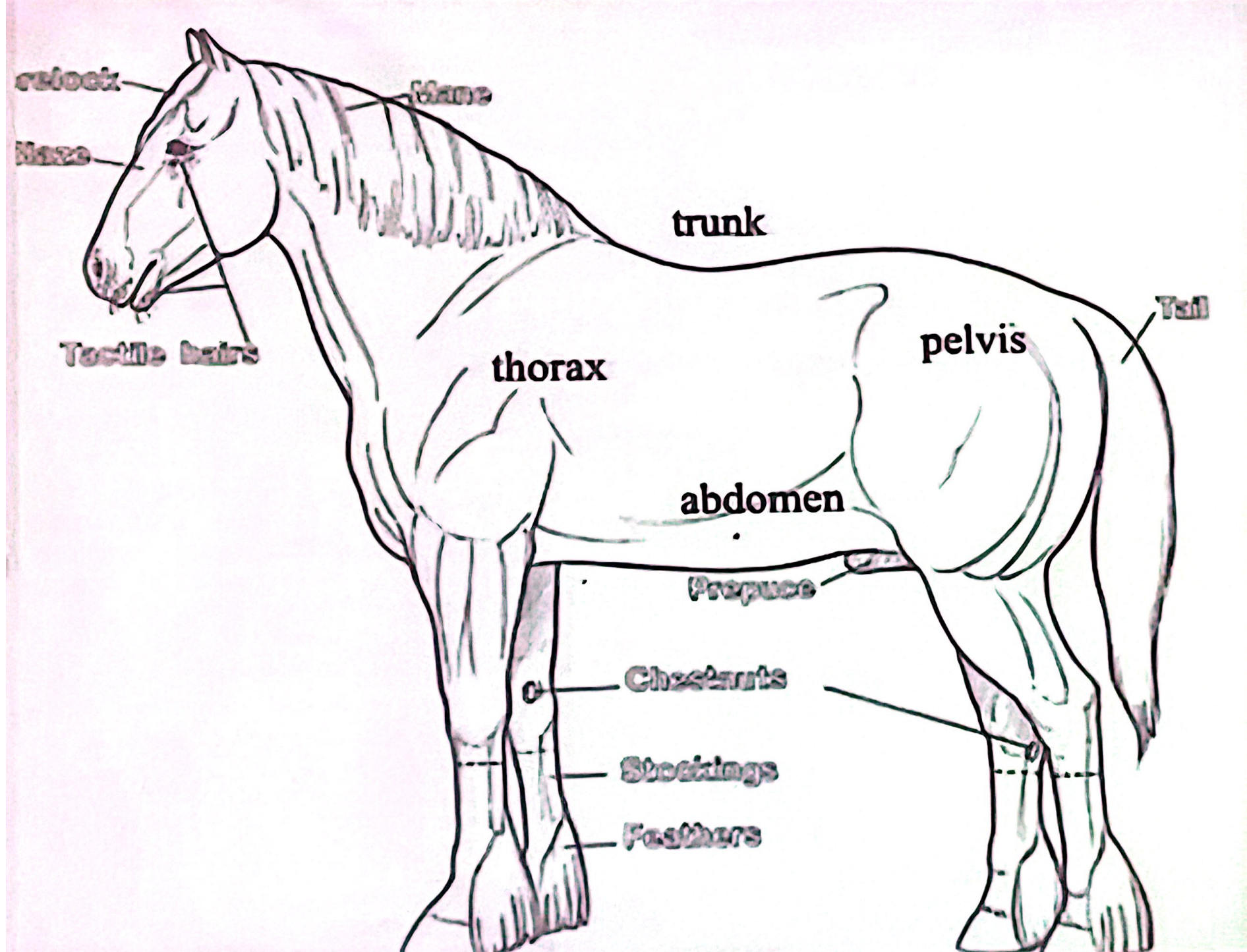
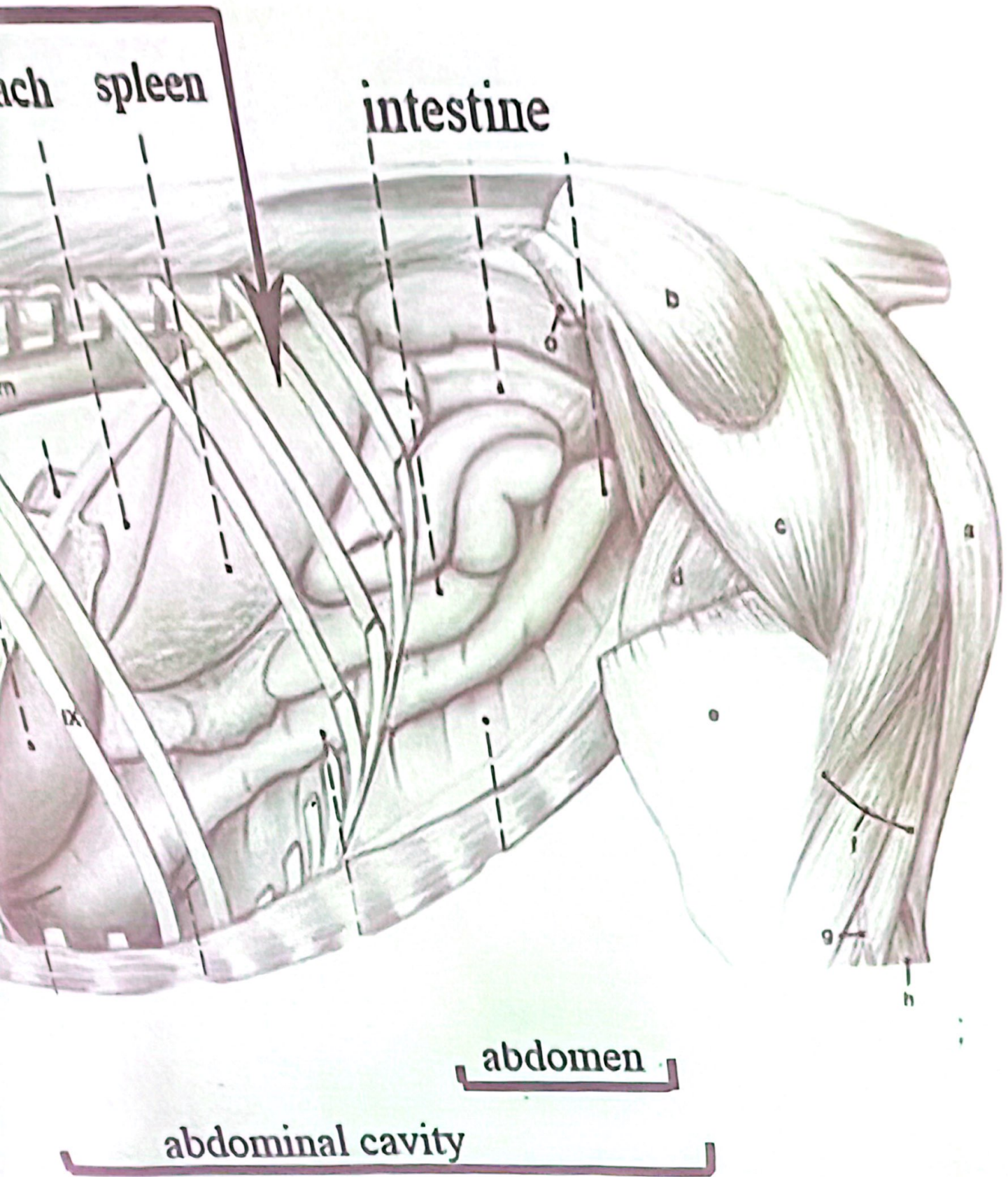




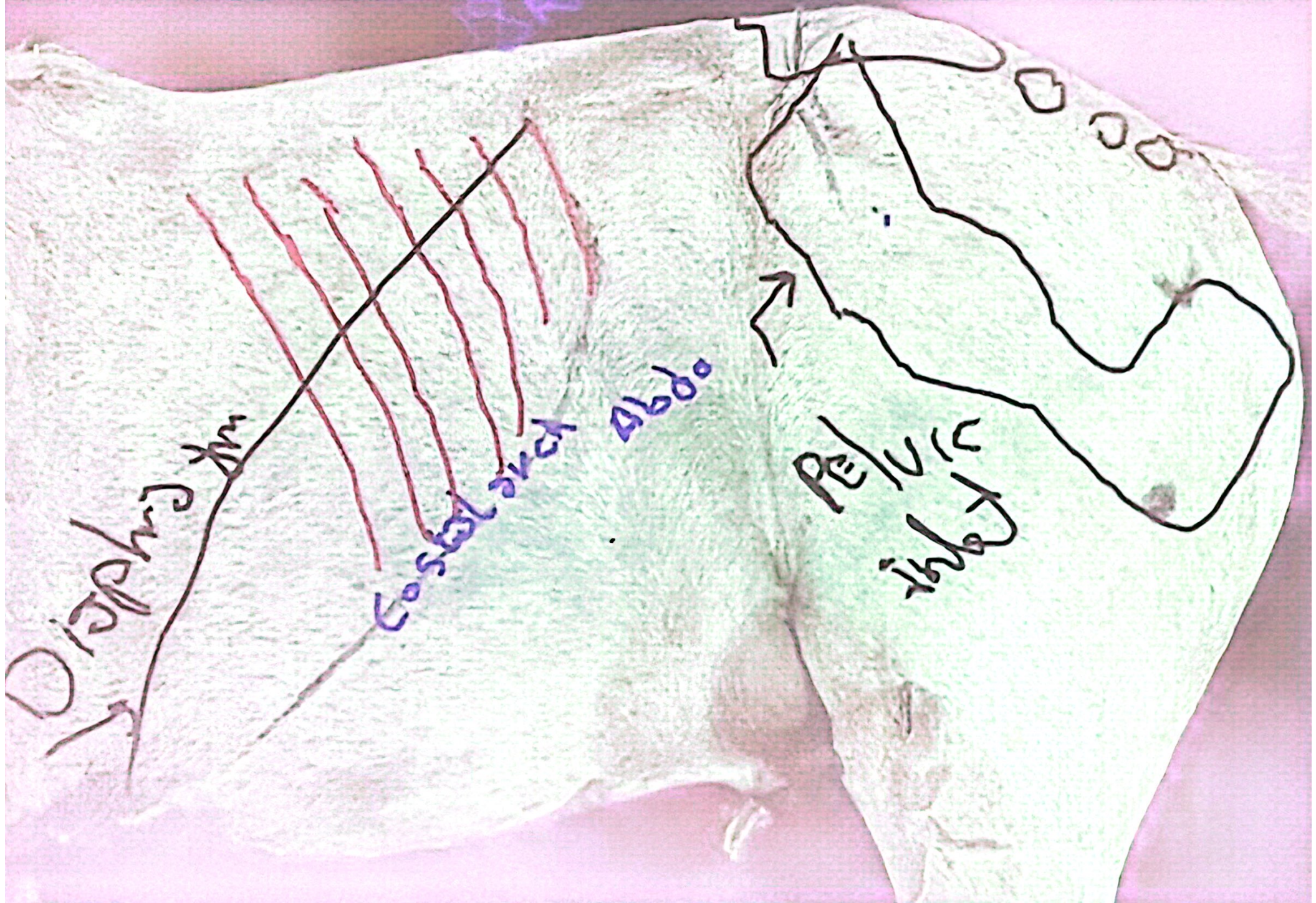
c part of the
cavity



abdominal cavity = abdomen + intrathoracic part of abdominal cavity

Abdominal cavity

- The abdomen is the part of the trunk that extends from the costal arch and the last rib to the terminal line (pelvic inlet) .This segment of the trunk contains the abdominal cavity, that appear ovoid in longitudinal section and circular in transverse section



- **Abdominal cavity** = cavity inside abdomen + intra thoracic part of abdominal cavity
-
- The boundaries of the **abdominal cavity** bounded **cranially**: by diaphragm
- **dorsally** :by dorsal abdominal wall
- **ventrally and laterally** by ventral and lateral abdominal wall
- and **caudally** it continues with pelvic cavity (**pelvic inlet**)

Dorsal abdominal wall consists of, skin - superficial and deep thoracolumbar fascia,
the **epi axial muscle** (ilicostalis, longissmus dorsi and multifidus) and the lumbar vertebrae.

Hypoaxial muscle: (quadratus lumborum, iliopsoas, psoas minor) and the iliac fascia

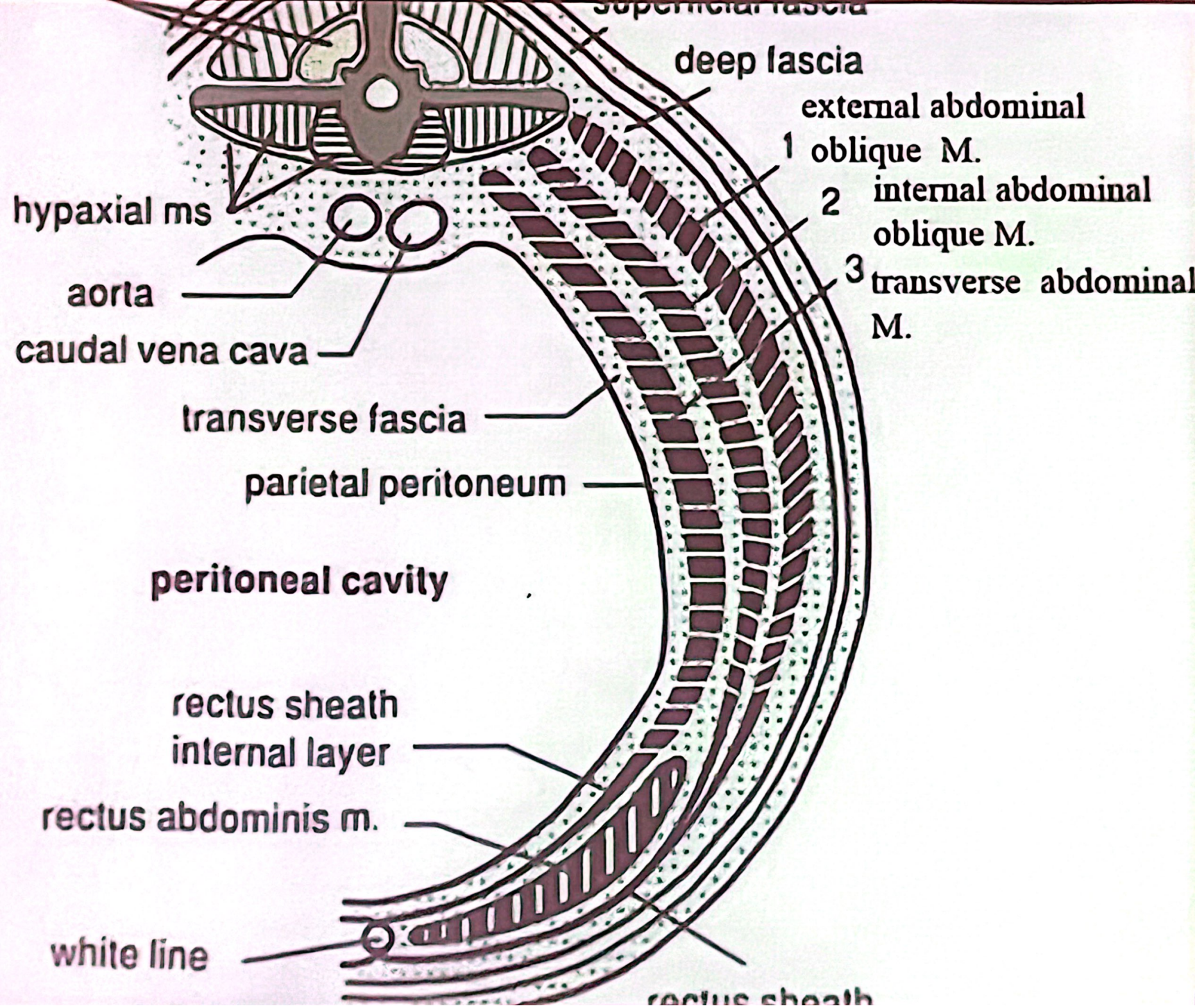
Lateral and ventral abdominal wall:
skin,

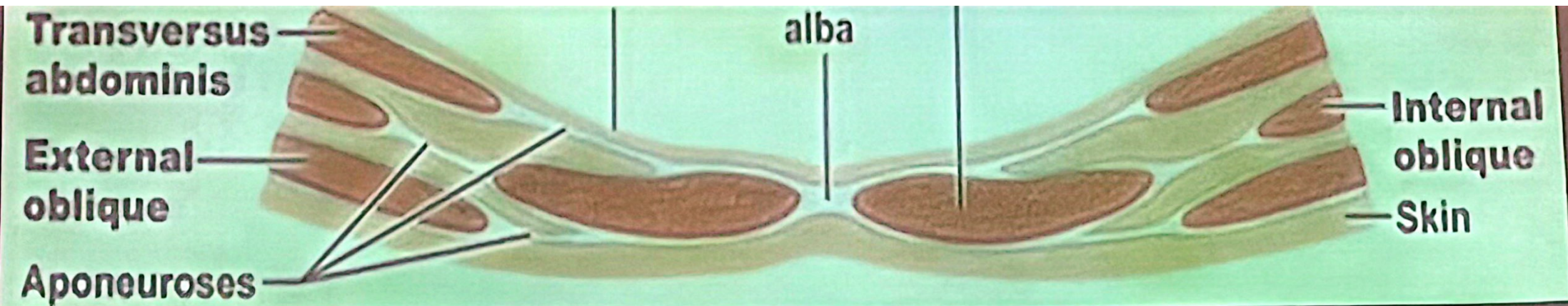
1-superficial and deep fascia ,

2- **abdominal muscle** (external and internal abdominal oblique muscles , **transverse** abdominal muscles and **rectus** abdominal muscles).

3-transverse fascia

4- and peritoneum





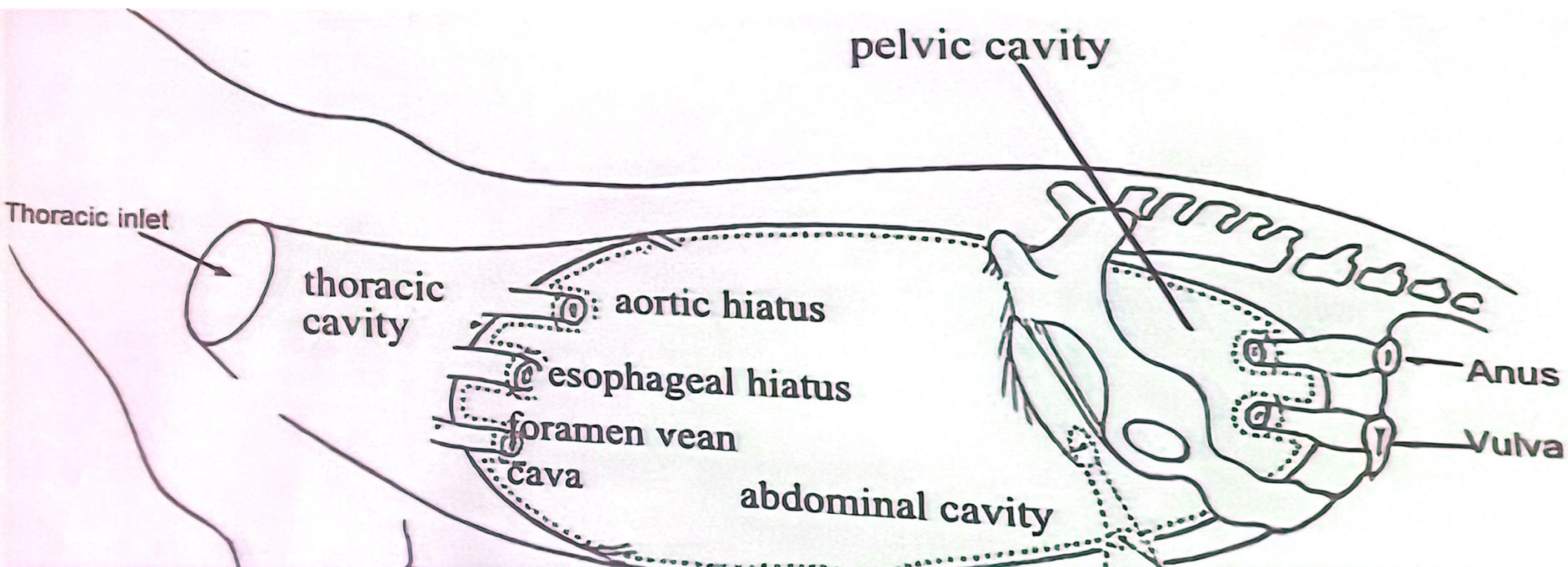
Rectus sheath : consist of two layers or lamina which enveloped the rectus muscles .

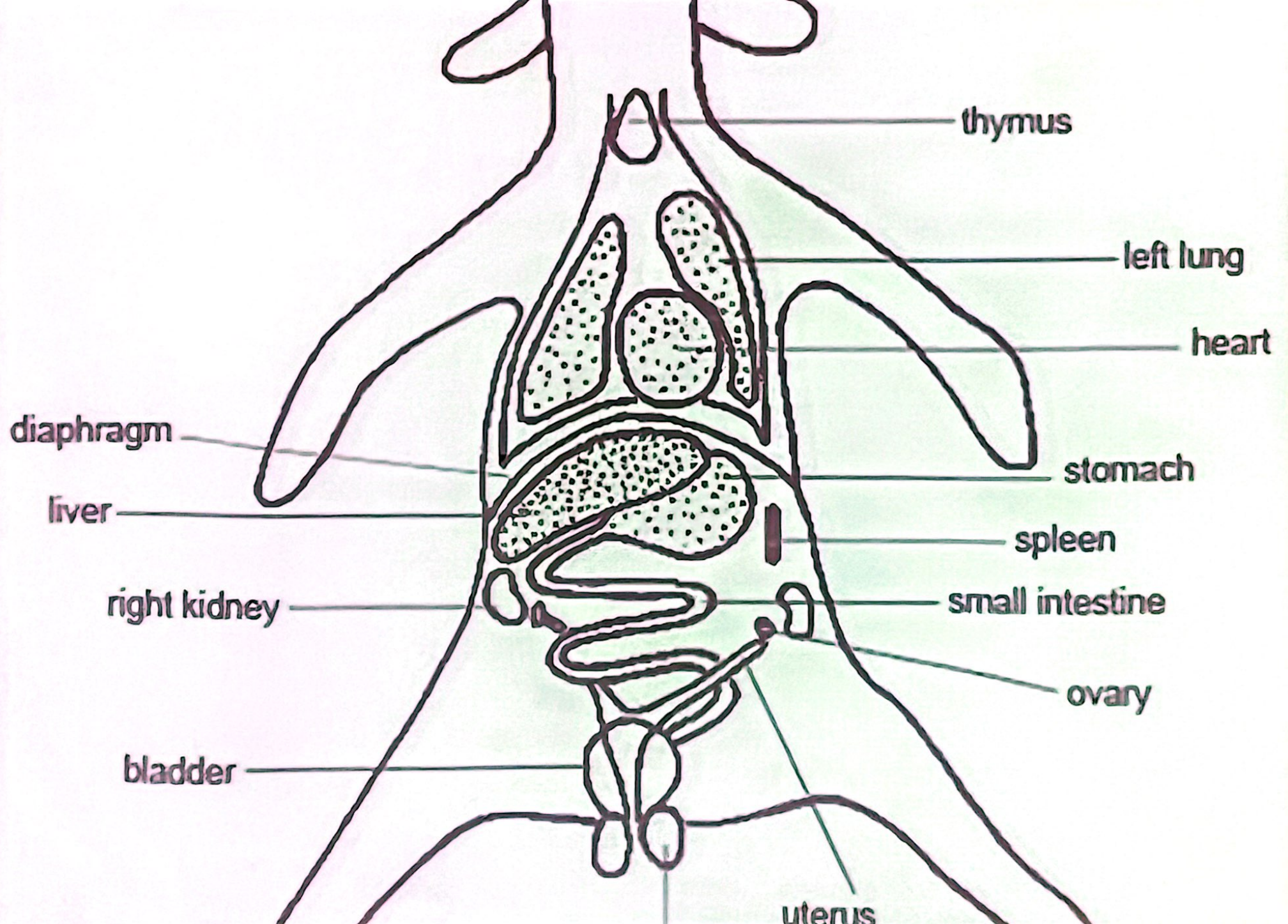
External lamina = union of aponeurosis of two muscles(external and internal abdominal muscles)**and**
Internal lamina= aponeurosis of transverse abdominal muscle

Linea alba it is a white fibrous line that extended at the midline of the abdomen from the xiphoid process to

openings :

- ❑ Aortic hiatus(Aorta, cisternachyli and vena azygus).
- ❑ Esophageal hiatus (esophagus, dorsal and ventral vagal nerve trunk and esophageal branch of left gastric) artery.
- ❑ Foramen vena cava (caudal vena cava passes).
- ❑ inguinal canal :Two in the inguinal area of the ventral abdominal wall
- ❑ umbilical canal. In fetus or newborn animal
- ❑ In male the peritoneal cavity is closed,
- in female, it is open to outside through the genital tract





-**Omentum**: This connecting fold passes from curvatures of the stomach to the organ of the abdomen

-**mesentery** is connecting fold, which attaches the intestine to the (dorsal wall of the abdomen).

-**Ligament**: are folds which passes between viscera other than parts of the digestive tube and attached it with the abdominal wall. Lateral of urinary bladder and coronary ligament of liver and broad ligament of uterus

- **The pelvic peritoneum**: is continuous cranially with abdominal cavity. it lines the cavity for variable distance

The peritoneum is thin serous membrane which lines the abdominal cavity and pelvic cavity (in part). In male is completely closed sac but in female it has two small opening these are the abdominal opening of the uterine tube or fallopian tube which there other ends communicates with the uterus

Parietal peritoneal = lined inner abdominal wall

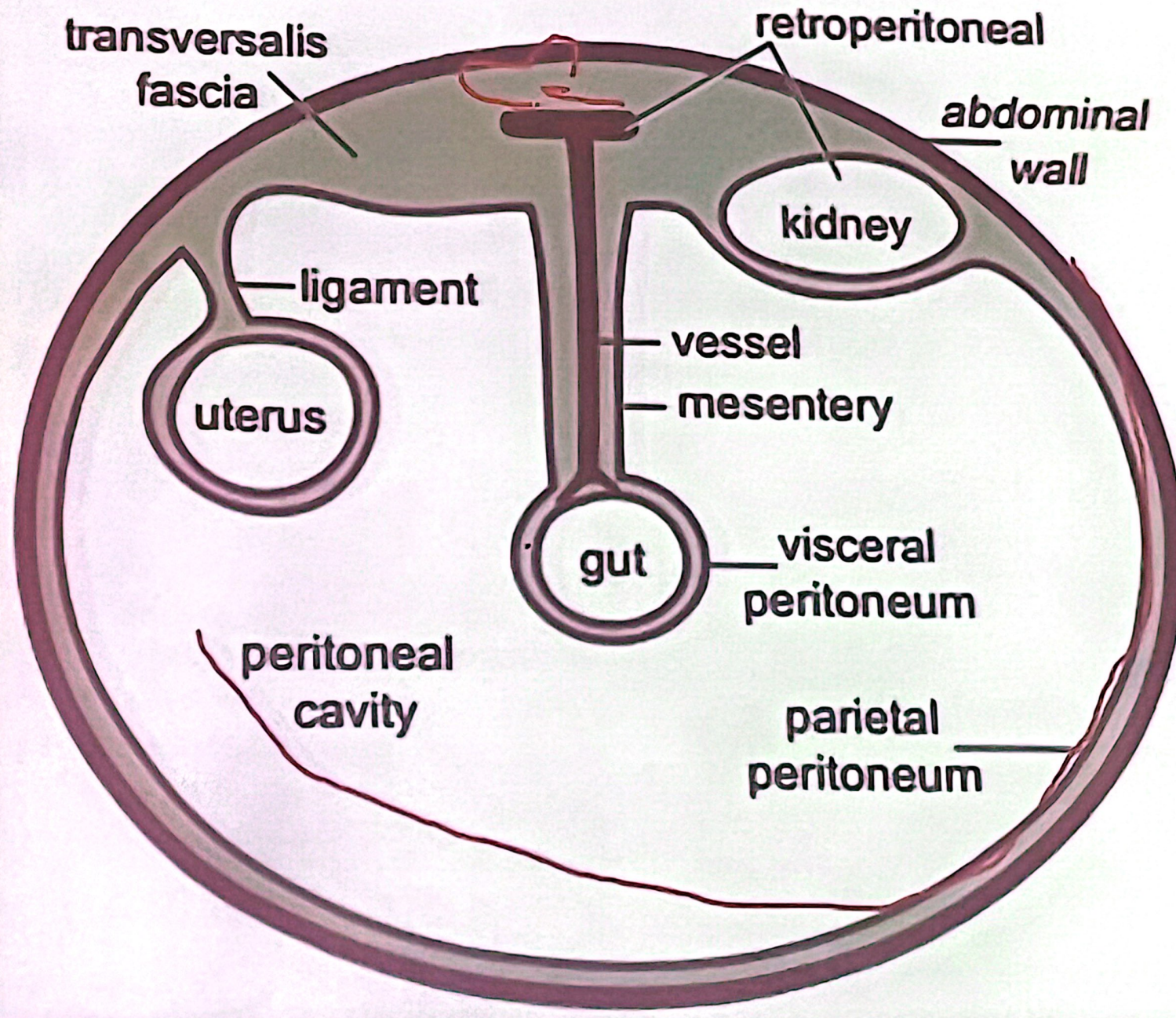
Visceral peritoneal = cover the viscera like stomach

Great peritoneal cavity = cavity between the visceral and parietal peritoneum

Peritoneal fluid = present at peritoneum cavity facilitate movement of organs or reduce friction between them.

Peritoneal folds = omentum + mesentery + ligament

Peritoneal Cavity



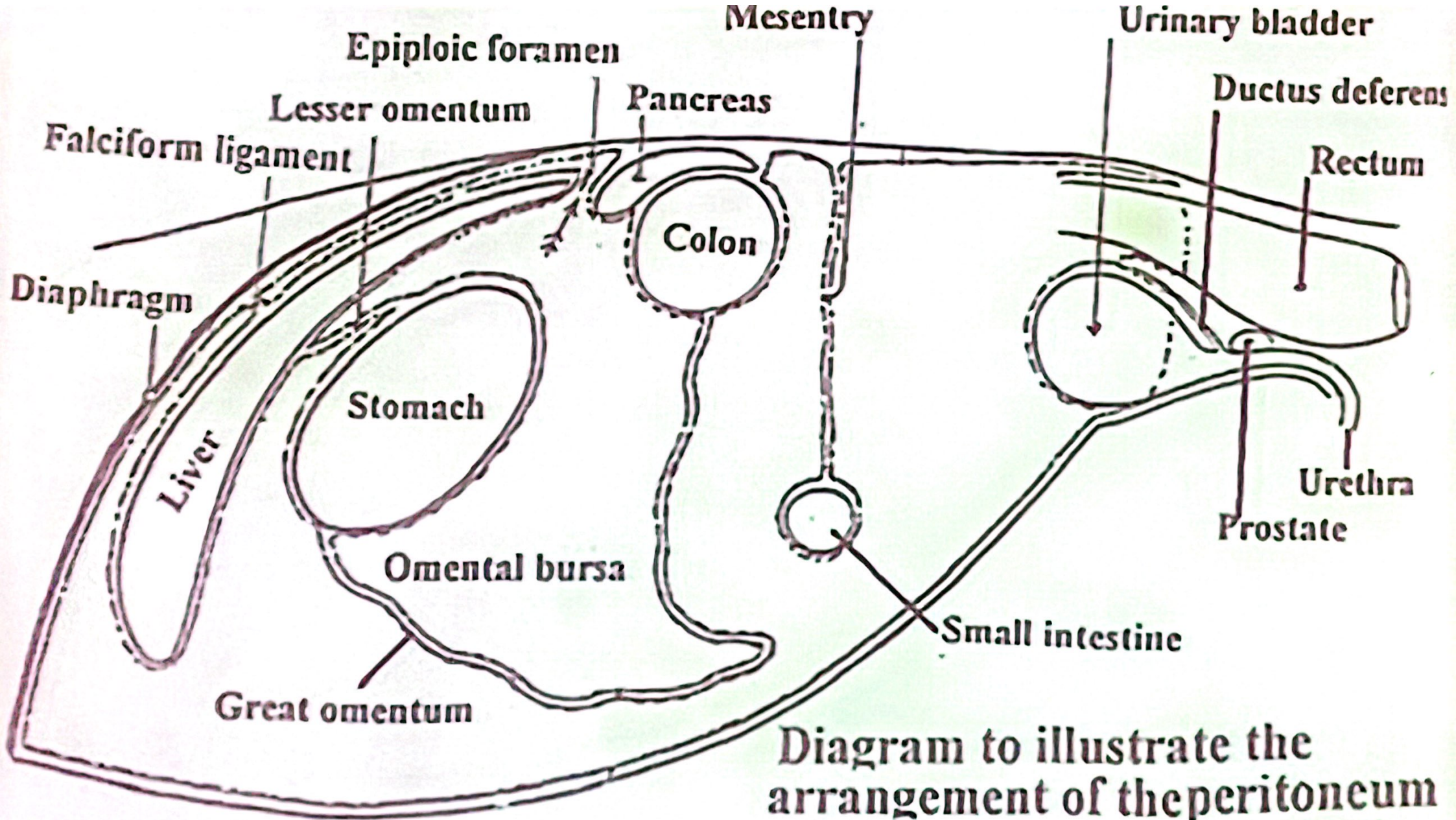


Diagram to illustrate the arrangement of the peritoneum in median longitudinal section of the abdomen and pelvis.

The peritoneum is thin serous membrane which lines the abdominal cavity and pelvic cavity (in part). In male is completely closed sac but in female it has two small opening these are the abdominal opening of the uterine tube or fallopian tube which there other ends communicates with the uterus

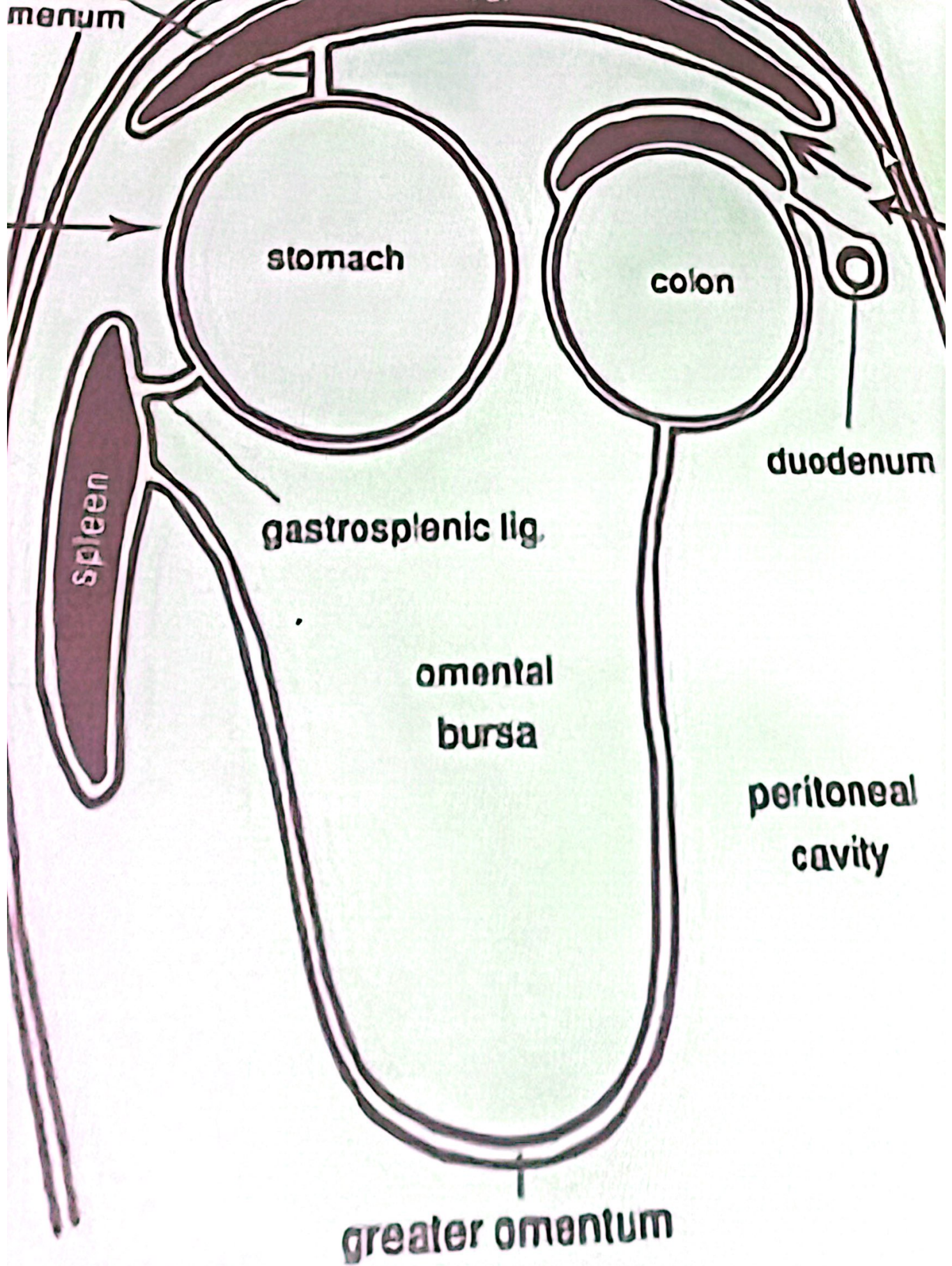
Parietal peritoneal = lined inner abdominal wall

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Great peritoneal cavity = cavity between the visceral and parietal peritoneum

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Peritoneal folds = omentum + mesentery + ligament



The epiploic foramen:

is a slit like opening to the right of the median plane lead to omental bursa : entrance of omental bursa
foramen is formed by caudate process of liver and caudal vena cava, the portal vein and pancreas and hepato-duodenal ligament□

Omental bursa (lesser peritoneal sac):

hollow space that is formed by the greater and lesser omentum and its adjacent organs. It communicates with the greater sac via the epiploic foramen

Consists of vestibule and number of recesses (dorsal), caudal and spleen

Pelvic cavity

It is the third cavity of the trunk which is enclosed in the pelvis. The caudal one. Open at both ends. Cranially (pelvic inlet) and caudally (pelvic outlet). Pelvic inlet communicates with abdominal cavity. The pelvic outlet is closed by the anus (male) and vulva (female) and pelvic diaphragm to form perineum.

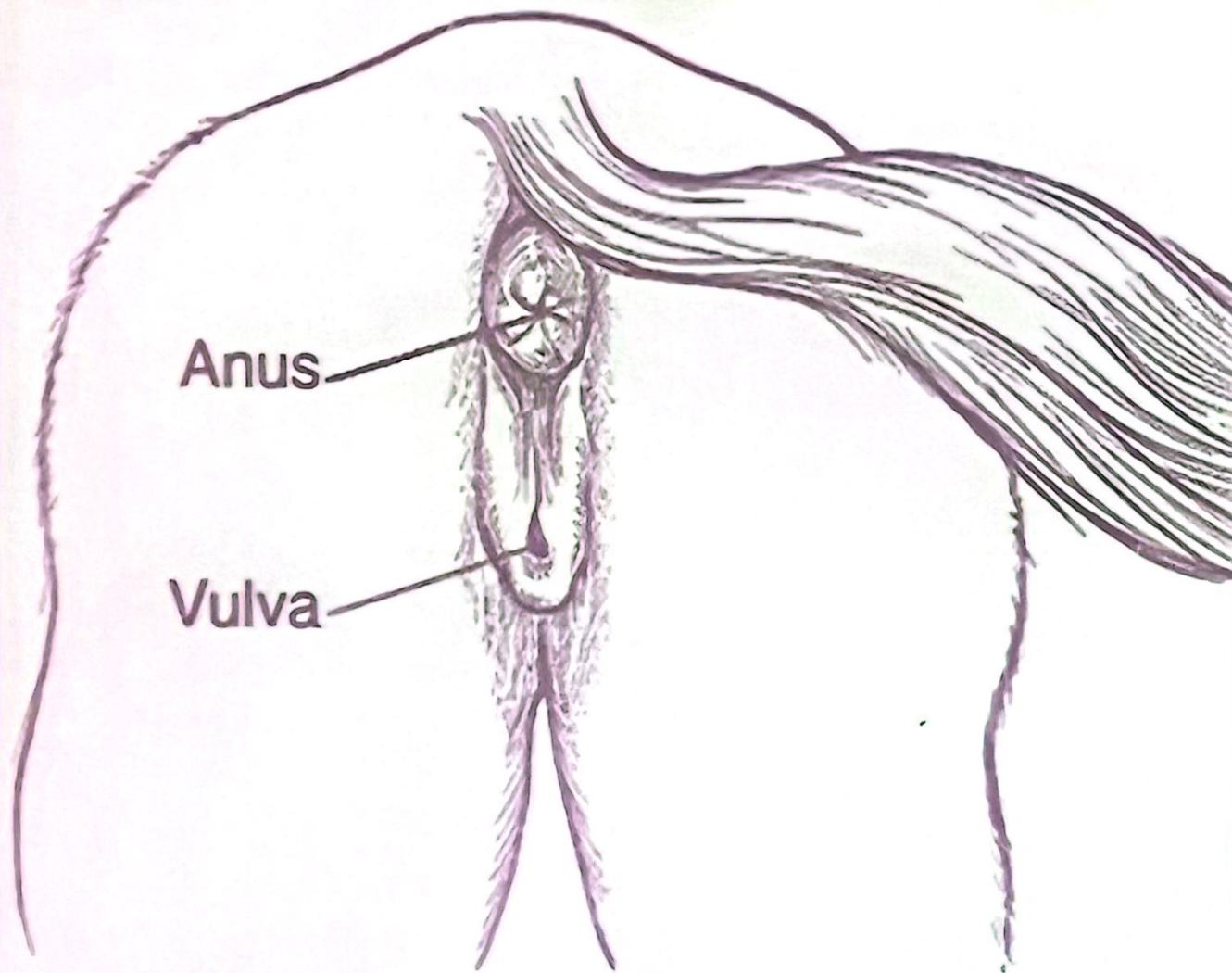
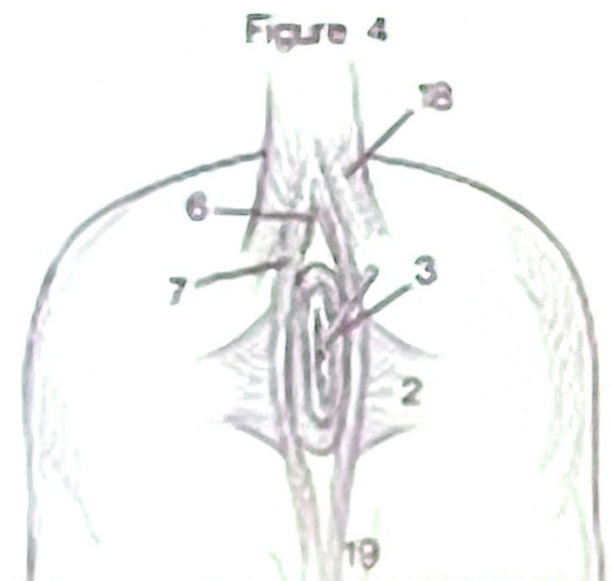
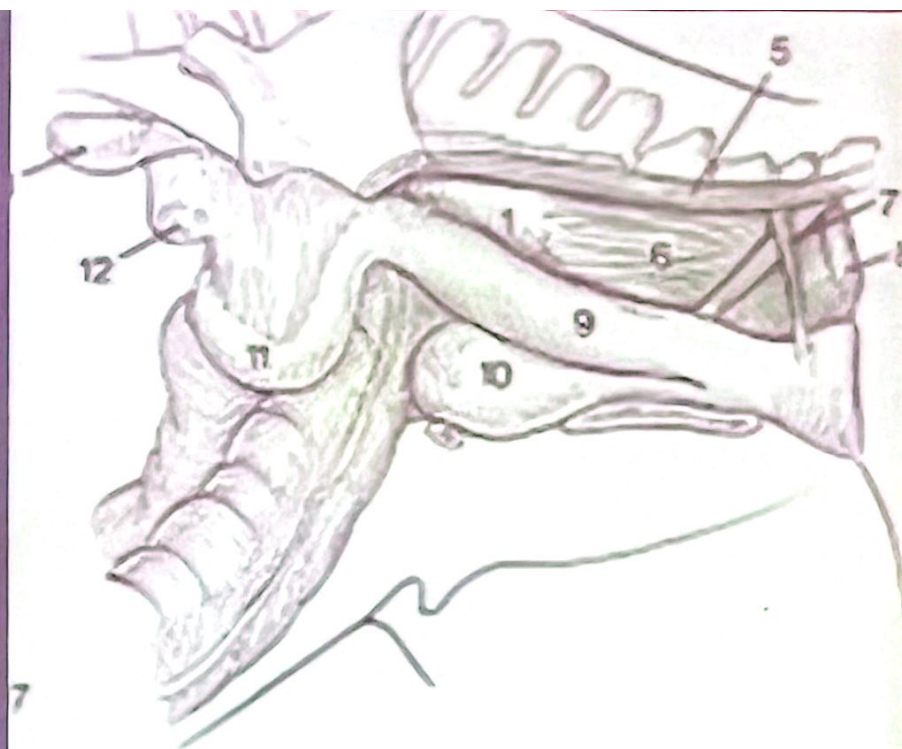


Figure 26-7. External genitalia of the mare.



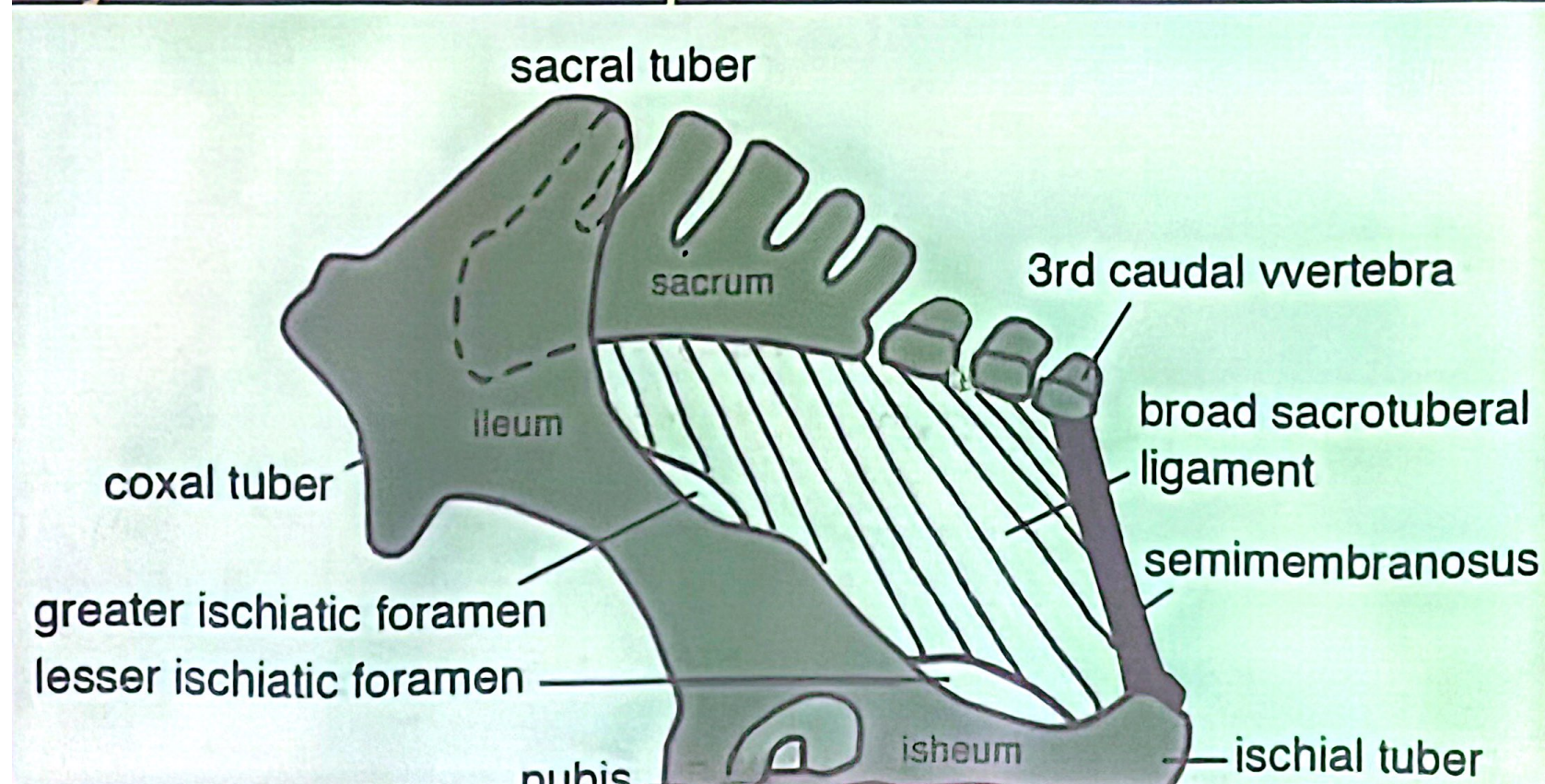
lly: by the sacrum and

first 3 or 4 caudal vertebrae

ly: ilium , broad sacrotuberal ligament

emimembranosus muscle

ally: ischium and pubis



sacrum

wing of ilium

base of sacrum

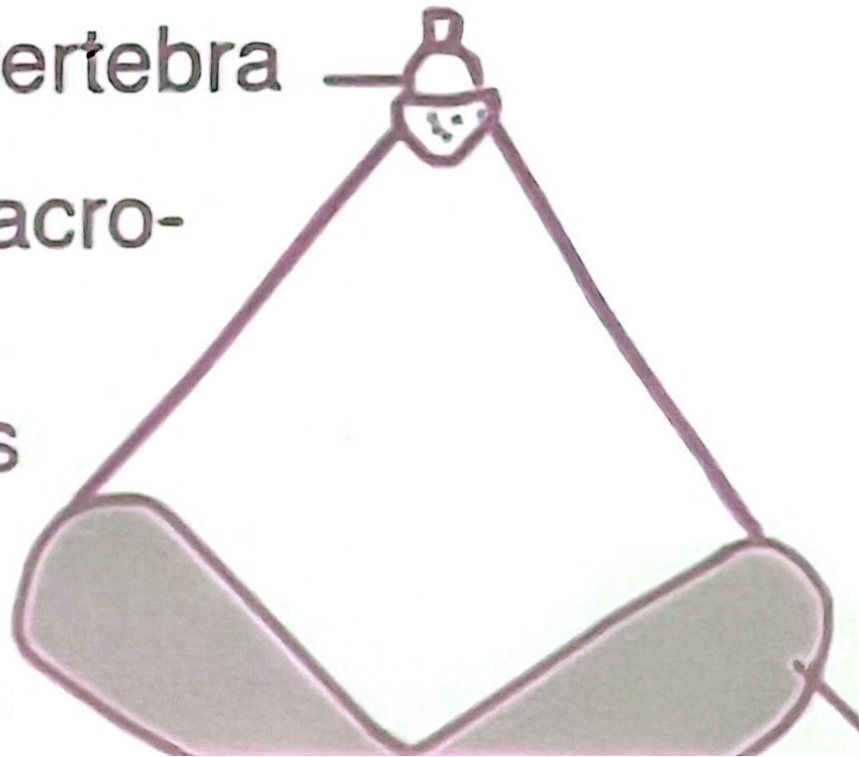
pelvic inlet

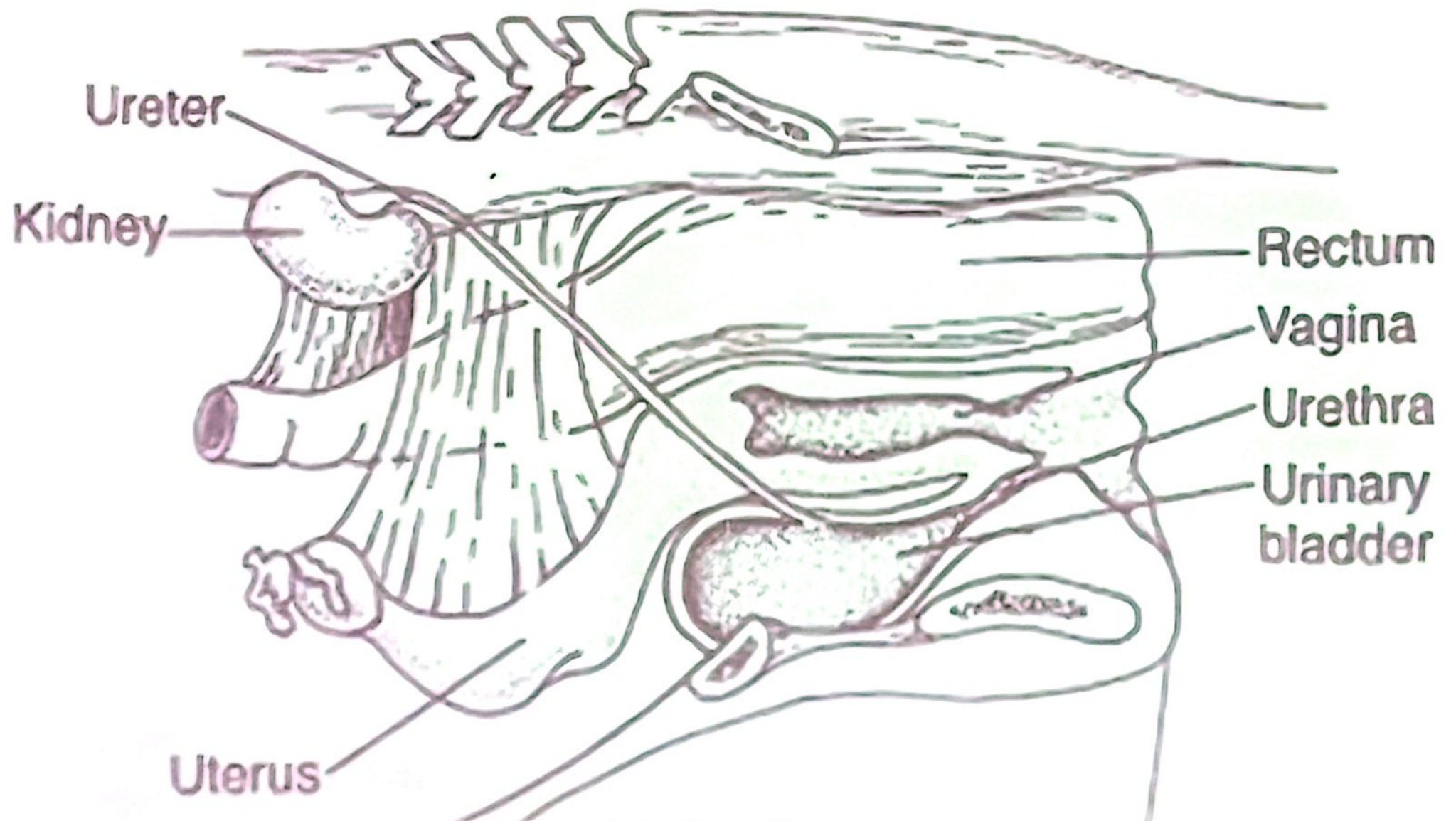
body of ilium

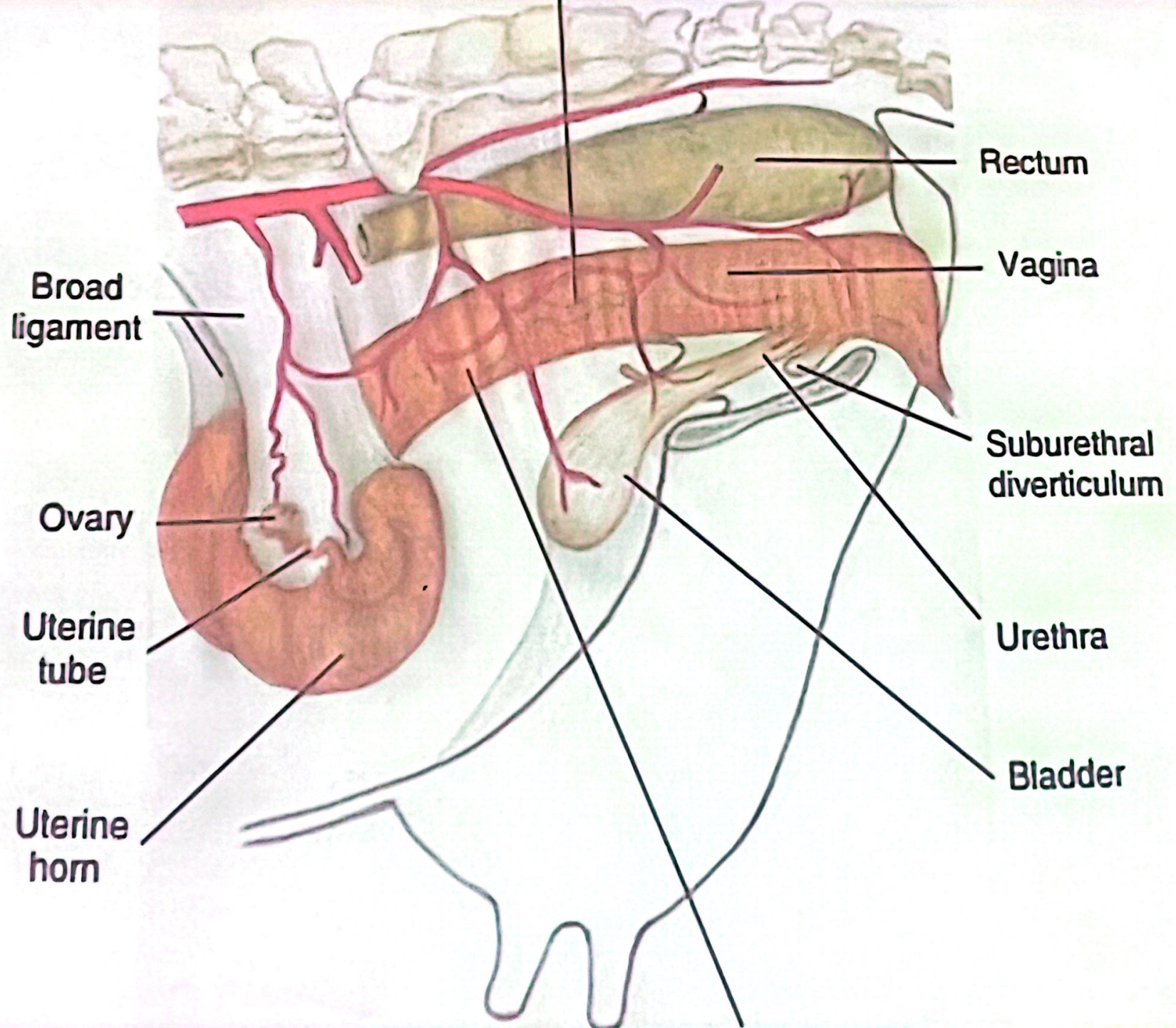
pectin of pubis

3rd caudal vertebra

caudal border of sacro-
tuberal ligament &
semimembranosus





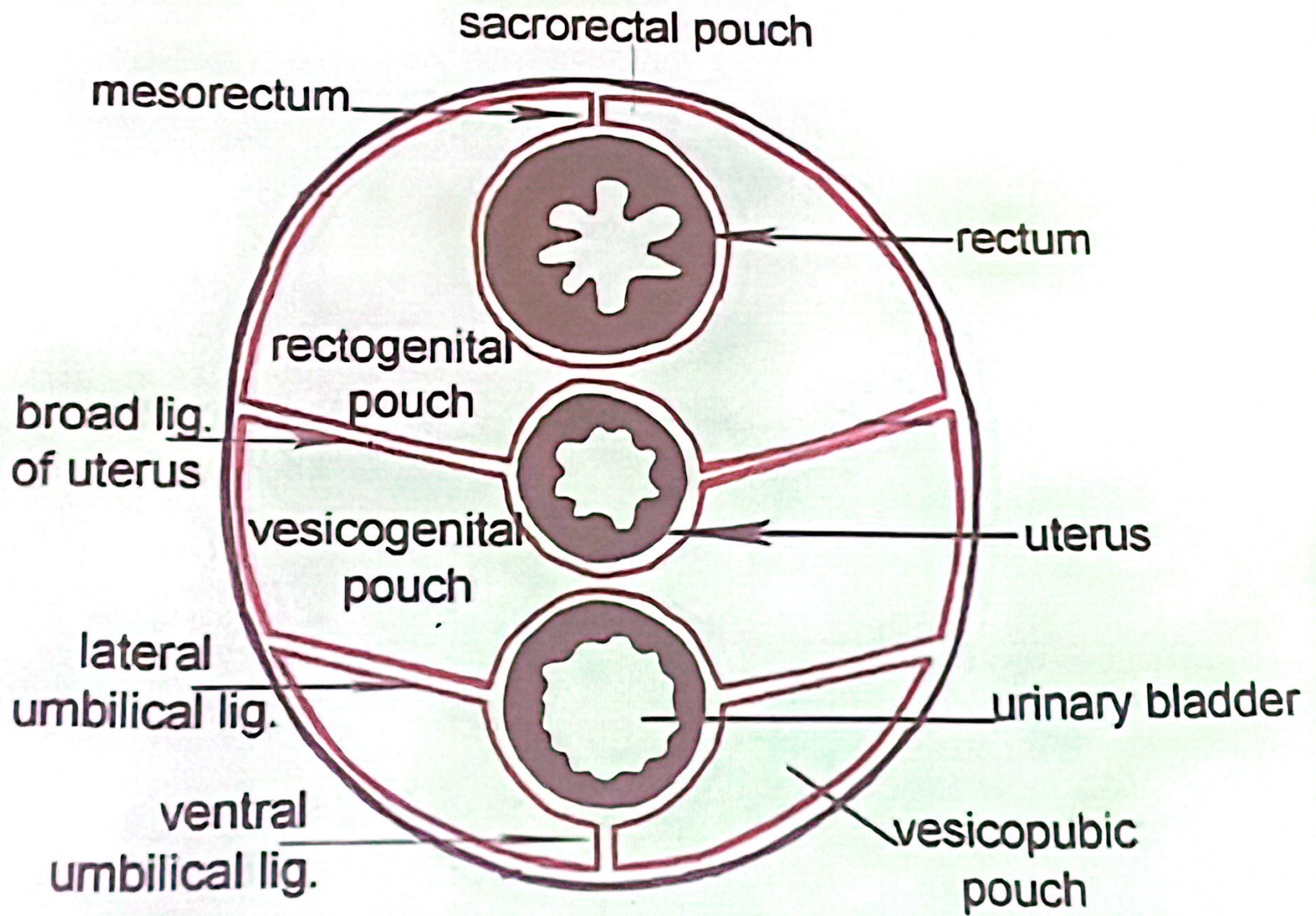


abdominal cavity. it lines the cavity for a variable distance is then reflected onto the pelvic organ forming pouches, while the caudal part is retroperitoneal and is covered only by fascia

Pouches: sacrorectal p., Rectogenital p., vesicogenital and vesicopubic pouches

Notice the following peritoneal folds:

- 1. Mesorectum attaching completely rectum**
- 2. Genital fold attaching vas deferens (male)**
- 3. Broad ligament attaching uterus (female)**
- 4. Lateral ligament of urinary bladder**



cross section in pelvic cavity of female peritoneal part